Cholesky

1

$x \in [0, L]$

$\log E(x)$

Gaussian Random Field, GRF FEM FNO

$x_0, x_1, \ldots, x_{n-1}$

$$\boldsymbol{y} = \begin{bmatrix} \log E(x_0) \\ \log E(x_1) \\ \vdots \\ \log E(x_{n-1}) \end{bmatrix}.$$

$\log E(x)$

$$\log E(x) \sim \mathcal{GP}\big(\mu,\ k(x, x')\big)\,,$$

$\boldsymbol{y}$

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{K})\,, \quad \boldsymbol{\mu} = \mu\mathbf{1}, \quad K_{ij} = k(x_i, x_j).$$

$\boldsymbol{K}$

2

x_i, x_j

$$K_{ij} = \mathrm{Cov}[\log E(x_i), \log E(x_j)]\,.$$

- $K_{ii} = \mathrm{Var}[\log E(x_i)]$ σ^2
- K_{ij} $i \neq j$ K_{ij}

  ℓ σ

$$k_{\exp}(x, x') = \sigma^2 \exp\left(-\frac{|x - x'|}{\ell}\right), \qquad k_{\mathrm{rbf}}(x, x') = \sigma^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right).$$

3 $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{K})$

$$\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}),$$

$\boldsymbol{A}$

$$\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{K},$$

$$\boldsymbol{y} = \boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z},$$

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{K}).$$

$\mathbb{E}[\boldsymbol{z}] = \mathbf{0}\ \mathrm{Cov}(\boldsymbol{z}) = \boldsymbol{I}$

$$\mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{\mu} + \boldsymbol{A}\boldsymbol{z}] = \boldsymbol{\mu},$$

$$\mathrm{Cov}(\boldsymbol{y}) = \mathrm{Cov}(\boldsymbol{A}\boldsymbol{z}) = \boldsymbol{A}\,\mathrm{Cov}(\boldsymbol{z})\,\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{I}\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{K}.$$

$\boldsymbol{y}$

4 Cholesky

$\boldsymbol{K}$ Cholesky

$$\boldsymbol{K} = \boldsymbol{L}\boldsymbol{L}^\top,$$

$\boldsymbol{L}$ $\boldsymbol{A} = \boldsymbol{L}$

1. $K_{ij} = k(x_i, x_j)$
2. $\boldsymbol{K} \leftarrow \boldsymbol{K} + \varepsilon \boldsymbol{I}\ \varepsilon$ 10^{-8}
3. Cholesky $\boldsymbol{K} = \boldsymbol{L}\boldsymbol{L}^\top$
4. $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$
5. $\boldsymbol{y} = \boldsymbol{\mu} + \boldsymbol{L}\boldsymbol{z}$

5 “jitter” $\boldsymbol{K} + \varepsilon \boldsymbol{I}$

$\boldsymbol{K}$

- ℓ
- Cholesky

$\varepsilon \boldsymbol{I}$

6 “ ”

$\boldsymbol{z}$ “ ” $\boldsymbol{L}$ “ ”

7

$\boldsymbol{y} = \log \boldsymbol{E}$

$$E(x_i) = \exp(y_i).$$

E $\log E$ μ, σ